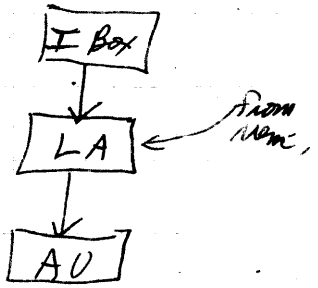
Meeting with Engineers Tues Jan 7, 1958
Sigma 9 Basic Status

(Bahnsen)

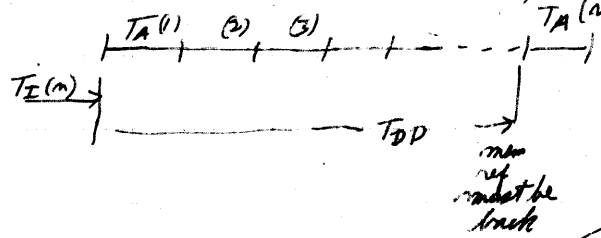


Consider 2 cases: (2) I faster than AU, (1) AU faster than I

(2) $\overline{T_I} \leq \overline{T_A}$ Case:

AU should run continuously;

for short instructions need pile up in LA. n levels

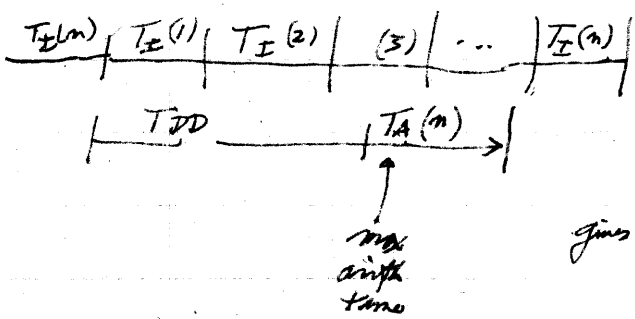


use a min?

So $(N-1)T_A \geq T_I + T_{DD}$ starting from LA req.

or $NT_A \geq T_I + T_{DD}$ if filling + emptying at same time.

(1) $\overline{T_A} \leq \overline{T_I}$ Case:



gives $(N-1)T_I \geq T_{DD} + T_{A\,max}$

optimum : equal times —

$\frac{T_{DI}}{2} \leq T_I$ for 1 buffer register,

T_{DD} access time from 2 μs mem = 0.6 μsec { incl. 0.4 address comparison

(incl. modif. of addr.)

to get slot	0.25	Data 2 final
50' cable	0.075	
Read	0.8	0.4 ← only change from 0.5 mem.
	0.2	
50' cable	0.075	
Set register	0.05	
	1.05	
Hamming check	×6	
	2.65 μs = T_{DI}	2.25 μs

for LA address comp. is removed (already done) −0.4 = 2.25 μs; −.4 = 1.85 μs

So : T_{DD} = 2.25 (2.0 μs Mem)

T_{DD} = 1.85 (0.5 μs Mem)

Instruction timing

0.3 μs	transfer instr to op register $Y_L \rightarrow Z_L$
0.3 μs	XSR → XR $Y_R \rightarrow Z_R$
0.6	add
0.6	addr. comparison via LA + transfer to LA (also start 0.3 XSR. 0.3 add overlap)
0.3	add
0.6	add comp. & transfer
0.2	finish I_3 + I_4 check

2.9 μs for 2 instrs. = 1.45 μs per half word instr.

So T_I = 1.5 μs

average $\overline{TA}$

6 ops @	0.5 μs	= 3.0
6 op @	1.0 μs	= 6.0
3 op @	2.5 μs	= 7.5
1 op @	7.5 μs	= 7.5
		24/16 = 1.5 μsec.

TA_{min} = 0.75 μs ← using one of 1st 2 times

Then $\frac{(n-1)}{\text{or } n} = \frac{2.2+1.5}{.75}$ = 5.0

using average 1.5 $(n-1) = 3.0$

for long ops. $(n-1) = 8.0$

---

case 2:

$\overline{TA}$ =

6 ops @ .2	1.2
6 ops @ .6	3.6
3 ops @ 1.2	3.6
1 op @ 1.8	1.8
	10.2/16 = 0.63 μs ← TA = $\overline{TI}$

$\frac{N-1}{\text{or } N} = \frac{2.2+0.63}{0.4}$ = 7.

TA_{ave} $\frac{2.2+.63}{0.63}$ = 4.5

TA_{max} $\frac{2.2+1.8}{.63}$ = 6.3

Still haven't taken into account in TA's

(1) geometrical layout
(2) clock cycle statistics

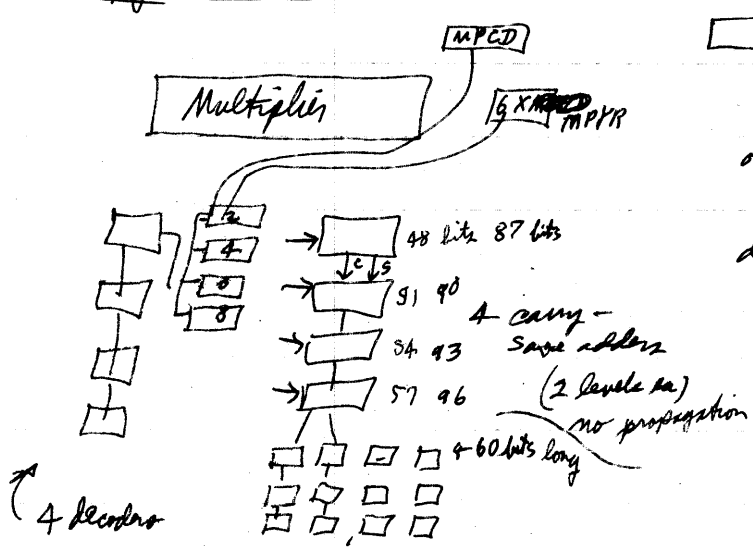
must assume better than 22 mμs to get these!

based on 22 mμs stages, (end of life)

18 mμs might be typical.

10 fast.

Mpy Scheme:



adder

add 2, 4, 6, 8 or sub each time determined by looking at nxt high order bit

← 188 "half adders" equiv.

~20,000 Transistors

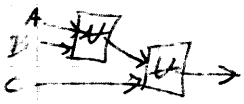
300 Transistors per gate
200 per adder.

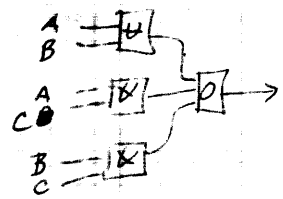
carry propagate adder (already in machine)

Repeat 4 cycles put into main adder — accumulate carries.

Total time:

bring mpy
normalize if necc
bring mpcand



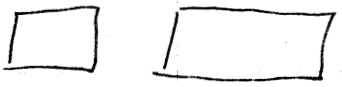


20	levels to get 0 x Mpcand
4	levels decode
8	" to go thru 4 adders.
10	shift back + redecode
8	
10	
8	
10	
8	
22	to gate to main adder
6	to normalize
5	exponent correction
119	

x22 = 2,618 μs

$$\frac{n}{K3}\left(K\sqrt{\;}+\sqrt{\;}\right)$$

Add Scheme:



bits are compared on carries
"look ahead" on carries.

Exp. difference,	13	(risd. getting no. from AA)
Mantissa shifts (0-8)	10	partly overlapped
Mantissa arith (48 bit sum no complement)	13	straight add with pre-compl.
Shift Normalizing again (0-8)	10	
Store in AB	2	
	48	x22 = 1.0 μs

recompl 13 more if necessary.

---

compare 704			present B+S		"old"	B	
6	24	144	.3	1.8	.2	3	18
6	84	504	1.0	6.0	.6	4	24
3	204	612	2.5	7.5	1.2	12	36
1	216	216	7.5	7.5	1.8	30	30
		1476		22.8	~~2.~~ 10.2		108

64
x.6 = 40.

50x
Machine
now
not
100x

Cost:
704

Speed	1	12	50
Cost	40K	120K	250K

print B
4
4
20
30